

Westravelle Analytical Ltd, a Tarnovell company

XA-5 suspension: approved analytical report

ICH chamber programme

Report no. E-R-2024-226-01

Project: WA/2022/226 | LIMS: WST-22-007025

Client: Xarnelvune Pharmaceuticals

Protocol interval: 08 Aug 2022 to 08 Feb 2024

Report date: 10 Feb 2024

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Status: Final, QA-approved

1 Summary and principal evidence

The resuspension assessment supported the dosing instructions and Xarnelvune added a shake-before-use instruction.

The following tables preserve the approved numerical data, units, result qualifiers and acceptance entries. They are read with batch-card identity, preparation and system suitability information. The scientific interpretation and client action are developed in the numbered sections.

Table 1. Approved stability results; units and qualifiers retained.

Condition	Timepoint (months)	Assay (% LC)	Imp A RRT 0.86 (%)	Imp B RRT 1.12 (%)	Total impurities (%)	Water KF (% w/w)	Result
25 °C/60% RH	0	100.1	0.07	<0.05	0.16	1.5	Complies
25 °C/60% RH	1	99.2	0.05	<0.05	0.17	1.3	Complies
25 °C/60% RH	3	99.1	0.07	0.06	0.24	2.5	Complies
25 °C/60% RH	6	99.2	0.09	0.06	0.25	1.2	Complies
25 °C/60% RH	9	99.6	0.12	0.08	0.25	0.9	Complies
25 °C/60% RH	12	99.2	0.15	0.11	0.32	2.5	Complies
25 °C/60% RH	18	98.9	0.20	0.09	0.43	2.1	Complies
30 °C/65% RH	0	100.1	0.07	<0.05	0.16	1.5	Complies
30 °C/65% RH	1	99.3	0.07	<0.05	0.25	1.9	Complies
30 °C/65% RH	3	99.3	0.07	0.07	0.24	1.0	Complies
30 °C/65% RH	6	99.1	0.14	0.09	0.34	1.3	Complies
30 °C/65% RH	9	98.5	0.18	0.10	0.40	2.0	Complies
30 °C/65% RH	12	99.2	0.23	0.14	0.51	1.9	Complies
30 °C/65% RH	18	98.5	0.30	0.18	0.57	1.0	Complies
40 °C/75% RH	0	100.1	0.07	<0.05	0.16	1.5	Complies
40 °C/75% RH	1	99.9	0.11	0.07	0.26	1.9	Complies
40 °C/75% RH	3	98.5	0.20	0.13	0.40	0.8	Complies
40 °C/75% RH	6	98.3	0.36	0.20	0.65	1.1	Complies

Specification: assay 95.0-105.0% LC; Imp A (RRT 0.86, specified) ≤0.50%; Imp B (RRT 1.12, specified) ≤0.30%; any unspecified impurity ≤0.20%; total impurities ≤2.0%

2 Report authority

This approved report is addressed to Xarnelvune Pharmaceuticals. Alice Penrose is responsible for scientific interpretation and release. Leon Wu maintains the sample and storage links; Saira Mahmood maintains the analytical method and system suitability links. The accession WST-22-007025 connects analytical preparations to batch cards, without replacing the client stability lot identity.

The signed report is the reporting copy. Sequence sheets, storage logs and original chromatographic files remain supporting material. An extract must retain the condition, timepoint, units and result qualifiers. Client decisions are distinguished from analytical findings in the conclusion.

3 Protocol objective

The work concerns XA-5 suspension supplied for Xarnelvune Pharmaceuticals, under WA/2022/226. Testing opened on 08 Aug 2022, the interim review was dated 09 May 2023, and analytical completion was recorded on 08 Feb 2024. The objective was to establish the analytical basis for the stated client decision.

Source material, exposure and preparation route belong to the result. A difference between analytical preparations is not automatically a difference between formulations. Investigative observations are discussed separately from accepted tabulated results. No extrapolated result is inserted into the reporting table.

4 Source-material traceability

The source chain begins with receipt and client stability lot description, continues through the batch card and withdrawal, and ends with the sequence sheet and approved result. Each preparation is linked to its source container. WST-22-007025 is the accession anchor for this chain.

The report does not create alternative stability lot numbers as free-text aliases. The source identifier remains on the batch card. A storage-condition label alone is insufficient evidence of sample identity. Source identity was reconciled before analytical trends were considered.

5 Receipt and custody

Product description, container state and accompanying instructions were entered against WST-22-007025. The custody trail distinguishes material reserved for the protocol from material removed for analysis. Handling after withdrawal is recorded because bench exposure can affect interpretation of a humidity comparison.

Retained material remains identifiable to its source container. A preparation remainder is not returned to that container. An investigative retained sample requires its exposure history to be established before comparison with scheduled material. The report assigns no unrecorded exposure history to a result.

6 Batch-set allocation

The batch set was managed so that source material, analytical preparations and reference preparations remained distinct. Client stability lot identities are carried on the batch cards. The sequence identifies the preparation used for each injection and its applicable exposure or challenge condition.

A fresh preparation examines extraction or handling, whereas reinjection examines an existing solution. These operations answer different questions and are not pooled merely because they share an accession. Subsamples from a container are not counted as independently manufactured stability lots.

7 Analytical responsibilities

Alice Penrose reviewed interpretation against the protocol objective. Leon Wu reconciled storage and withdrawal information. Saira Mahmood checked analytical sequences and system suitability disposition. Mark Campbell reviewed report-to-source links for the approved copy.

Approval does not replace contemporaneous analyst entries. Corrections remain attributable on their sequence sheets and original analytical files remain available. Scientific review asks whether the evidence supports the conclusion; quality review establishes faithful reporting and traceability without creating missing numerical evidence.

8 Suspension protocol

The programme ran from 08 Aug 2022 to 08 Feb 2024, with an interim review on 09 May 2023. Material for scheduled storage was distinguished from material handled for in-use assessment. Container history remained attached to preparations.

A settled suspension portion and a resuspended portion are not interchangeable samples. Handling evidence therefore mattered to the dosing interpretation alongside the chamber programme.

9 Withdrawal chronology

Leon Wu maintained the chamber and withdrawal trail under WST-22-007025. Condition and timepoint identify scheduled storage results. In-use handling is discussed separately rather than presented as another chamber condition.

Withdrawal ends continuous chamber residence for that sample. Subsequent handling remains on the sequence sheet, without presuming that an opened container is equivalent to an untouched one.

10 Resuspension preparation

Product handling state was recorded before representative sampling. Resuspension restored a usable dispersion for sampling and dosing without changing the assay calculation basis. Source links remained on sequence sheets.

The supported instruction is shake before use. No invented shake count, agitation duration or numerical recovery result is added. More detailed instructions would require their own controlled basis.

11 Assay preparation

Test solutions were prepared from identified source material using the protocol extraction and dilution route. Weighing, transfer, dissolution and dilution entries remain on the sequence sheet.

Preparations were protected from avoidable evaporation and contamination while awaiting injection.

Assay is expressed as percent label claim. The reference basis and dilution factors belong to the calculation. The report does not silently convert assay into a dry-weight result or correct it for water. Reported values are those in the approved numerical table.

12 Related-substances measurement

Related substances are reported as area percent under the applicable chromatographic method. Integration settings, peak assignment and approved qualifiers accompany the analytical sequence. Retention position assists comparison but is not, by itself, proof of chemical identity.

An investigative reprocessing remains linked to its reason and does not overwrite original integration history. A less-than entry is not replaced by zero. Individual impurities and total impurities are read in their stated units rather than treated as an assay deficit.

13 Reference-material control

Reference preparations are identified separately from test preparations. Their batch cards retain the material basis applicable to the calculation. Preparation and use are linked to the analytical sequence, so a visible reference peak is not alone evidence of a valid calibration.

A reference used for peak assignment has a different role from one used for assay calibration. An investigative addition to a sample is not automatically its quantitative result. Reference-only, sample-only and combined preparations are interpreted according to the question being investigated.

14 Sequence design

The sequence sheet separates blanks, reference, system suitability and test preparations. Injection order is retained so a sample can be related to its surrounding controls. Sequence completion does not itself establish analytical acceptance.

Blanks support assessment of interference and carry-over; bracketing reference preparations support response review. An unexpected signal is assessed against preparation history and injection position. A repeat sequence retains its own identity and link to the original. Rejected measurements are not averaged into accepted results.

15 System suitability

System suitability is a release gate for an analytical sequence. Method-defined checks are applied before sample calculations are accepted. Reference response and the separation relevant to impurity reporting are reviewed for the actual analytical question.

Detailed acceptance entries remain on the sequence sheets. This report does not invent numerical suitability values absent from the consolidated source. A sequence requiring investigation retains its initial disposition. Assay suitability cannot be presumed to establish suitability for a more demanding impurity assignment.

16 Calculations

Assay retains the percent label claim basis and related substances retain the area percent basis. Water, where tabulated, has its own mass basis. These quantities cannot be added or subtracted as interchangeable measurements. Rounding follows the controlled source table.

Missing measurements are not zero results and row status is not a numerical value. Comparisons require matched conditions and source links. The tables are reproduced from controlled project data, preserving qualifiers without manually introducing extra precision or derived result values.

17 Settling and sampling

Settling can change the composition of a sampled portion without establishing chemical degradation. The sampling question was therefore separated from assay and impurity trends. The chemical result table retains its original basis.

The assessment supported dosing instructions. It is not presented as a new numerical dose-delivery dataset, and dose-uniformity percentages are not inferred from assay.

18 In-use assessment

The assessment concerned resuspension in the intended handling context. Opening and sampling were exposure events with their own trail, rather than proof for every possible user handling pattern.

The practical outcome was a shake-before-use instruction. The report adds no microbiological conclusion or untested post-opening period; those questions require separate evidence.

19 Chemical interpretation

Assay remains percent label claim, impurities area percent, and water its separately stated mass basis. Interpretation requires the preparation state and source-container link.

Chemical stability is not a direct measure of ease of resuspension. Successful resuspension does not cancel a chemical result or status. Both forms of evidence are retained with their different meanings.

20 Dosing instruction

Xarnelvune Pharmaceuticals added a shake-before-use instruction. The action carries the resuspension assessment into practical use of XA-5 suspension without implying a formulation change.

The instruction should be available where a dose is prepared. An analytical preparation note alone would not communicate it to the user. This report records the decision without claiming to test every future wording.

21 In-use boundary

The final chamber interval is 18 months. This must not be read as a continuous established period after first opening. Chamber timepoint and in-use exposure answer different questions.

The report supports observed resuspension and the recorded instruction. It does not create an open-container shelf life, dosing-device compatibility result or microbial-preservation claim.

22 Implementation gate

The shake-before-use instruction remains connected to product handling requirements. A changed container, formulation or sampling arrangement requires reconsideration of the resuspension basis.

Analytical handover also retains sampling state. A portion drawn from an unspecified suspension state is difficult to compare with this protocol. Sample representativeness remains a continuing analytical control.

23 Data handling

Original chromatographic files, preparation entries and sequence sheets remain associated with WST-22-007025. The archive retains injection order, processing history and the relationship between a preparation and its reported result. A revised integration is distinguishable from the original.

Report assembly uses the approved project tables rather than a later working export. Investigative records are retained even when no reported result changes, because they explain the accepted interpretation. Additional analysis is treated as additional work rather than an undocumented alteration to this final report.

24 Interim and final review

The interim review on 09 May 2023 was considered against the evidence then available. The final interpretation incorporates work completed by 08 Feb 2024. An observation, an investigated explanation and a client decision remain distinguishable.

Completion closes the reporting interval under WA/2022/226. It does not extend exposure or create a shelf-life claim. A receiving laboratory needs the preparation basis and suitability requirements from the controlled method rather than having to infer executable instructions from this report.

25 Deviation-note disposition

Deviation notes distinguish a departure from the protocol from a product response during planned testing. An unfavourable result is not itself a procedural deviation. An acceptable result likewise does not cancel a documented handling departure.

Investigative measurements remain linked to the question prompting them. This report does not manufacture a chamber excursion, deviation number or repeat-test result absent from the source material. The disposition remains with the relevant sequence sheet and batch card; scientific consequences are discussed below.

26 Comparison limits

Source identity, preparation and exposure control the meaning of comparisons. Dissimilar conditions can inform interpretation but cannot be treated as replicate measurements. A condition-dependent response may support a handling instruction without establishing its chemical mechanism.

This report addresses XA-5 suspension within the completed interval. It does not certify a new supplier, changed manufacturing process, altered pack or untested environment. The recorded client action should retain the reason for the change in subsequent product documentation.

27 Implementation controls

Handover should preserve the approved method, reference basis, sample-preparation instructions and system suitability requirements. The receiving analyst must be able to link a result to its stability lot and exposure. A report table is insufficient as a stand-alone executable method.

A handling instruction must be reproducible and a sensitive method-preparation step must be explicit. Future changes should be assessed against the reason for the original decision, rather than accepted merely because the product name is unchanged. An altered configuration is not described as already tested.

28 Conclusion

The resuspension assessment supported dosing instructions and Xarnelvune Pharmaceuticals added a shake-before-use instruction. The chamber programme is reported through 18 months.

The conclusion concerns tested material and documented handling. It assigns neither a numerical dose-uniformity result nor an additional in-use expiry period.

29 Archive index

The reporting pair comprises this final report and XA5_PULLS_AND_SUITABILITY_RESULTS_Rev03.xlsx. Analytical completion was 08 Feb 2024; source records remain under WA/2022/226, accession WST-22-007025. Batch cards, exposure records and sequence sheets remain the authoritative detailed links.

The original project-year identifier is retained even where completion occurred in a later year. Branding follows the report date. A later commercial identity is not applied retrospectively to the reporting copy.

30 Batch-card retrieval

Retrieve the batch card using WST-22-007025, then confirm client stability lot and container before retrieving the sequence. Searching only by product name can return material from another exposure or preparation.

A result query should preserve the reported condition, timepoint and sequence identity. That trail distinguishes a material question from a processing question. This appendix is a retrieval guide, not a replacement batch card.

31 Sequence-sheet retrieval

The sequence sheet retains preparation entries, injection order, references and system suitability disposition. Retrieve the accepted original before comparing a reported value with an investigative reinjection.

Where reprocessing was used, retain original and revised processing links and the reason. A later working export cannot alone determine which preparation supplied the approved result. Qualifiers remain in any query response.

32 Result-query routing

Specify the report identifier, accession and affected row. Establish whether the concern is identity, exposure, preparation, integration, calculation or interpretation. Different questions require different records.

Alice Penrose handles interpretation, Leon Wu storage and material links, and Saira Mahmood method and sequence links. A response should preserve original units and limitations. A query does not authorize overwriting an approved result.

33 Controlled vocabulary

Protocol denotes the agreed programme; batch set the material grouping; sequence sheet the preparation and injection record; system suitability the analytical acceptance gate. Stability lot is source identity, not every new preparation.

A deviation note retains the issue and disposition. A release gate is an evidence-based decision point. Consistent vocabulary prevents storage, analytical acceptance and client action from being conflated in subsequent extracts.

34 Units and qualifiers

Assay is percent label claim and related substances area percent. Chamber temperature, where applicable, uses °C. Water KF retains the table's percent w/w basis. Headings belong to the result.

A qualified low-level entry is not an exact zero. Status remains with the row. No derived value from rounded data, retrospective water correction or new acceptance limit is introduced in this report.

35 Handover record

The handover retains the approved protocol, method, numerical tables and client action under WA/2022/226. It should also retain preparation and reference instructions needed to interpret the findings.

Transfer of a report is not proof of receiving-site performance. Additional work needs its own source links, objective and acceptance disposition. The conclusion here stays within the completed interval.

36 Reporting-copy closure

This final copy closes the interval stated in the title block. It retains the principal evidence, analytical basis and client action. Supporting records remain associated with WA/2022/226 and WST-22-007025.

A future amendment identifies the affected reporting element and source basis. Revised instructions or selections are not silently inserted into this dated report. The signed conclusion remains limited to material and work completed here.

Name	Role	Signature	Date
Alice Penrose	Head of Analytical Services — scientific approval	<u>A Penrose</u>	10 Feb 2024
Leon Wu	Stability Coordinator — material traceability	<u>L Wu</u>	10 Feb 2024
Saira Mahmood	QC Method Validation Lead — analytical review	<u>S Mahmood</u>	10 Feb 2024
Mark Campbell	QA Officer — reporting-copy approval	<u>M Campbell</u>	10 Feb 2024